

ALEX ABABU

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EDUCATION

MS in Artificial Intelligence in Business, Arizona State University, Tempe, AZ May 2026 | 4.0 GPA | Graduated with Distinction
BS in Computer Science, Unity University Nov 2021 | 3.82 GPA | Graduated with Distinction
Honors: Mastercard Foundation Scholar | Beta Gamma Sigma Honor Society (Top 10% of MS graduates)

EXPERIENCE

AI Acceleration Analyst, Arizona State University, Tempe, AZ Aug 2025 – Present

- Built a multi-modal evaluation platform (FastAPI, React, AWS ECS/Fargate, S3, DynamoDB) using Whisper for video and audio transcription and a two-stage LLM-as-judge pipeline for automated rubric scoring against custom rubrics; deployed via AWS CDK and GitHub Actions CI/CD
- Built a Slack-integrated RAG assistant on an event-driven AWS backend (Lambda, API Gateway, SQS, DynamoDB) with multi-tenant routing for asynchronous LLM support to ASU students and faculty
- Engineered an MCP-based persistent memory tool enabling context sharing across Claude, GPT, and Gemini; authored analytical SQL on Redshift and automated schema drift detection between Redshift and DynamoDB with Python

Software Engineer, Freelance Jan 2024 – Aug 2025

- Built a vehicle inventory management app (Python, OpenCV, CNN-based OCR) that uses camera-based license plate recognition to automate vehicle check-in/check-out tracking and reporting; delivered end-to-end to client specifications
- Engineered a real estate listing mobile app using Flutter with property upload, search, filtering, and in-app chat; implemented GraphQL backend and Firebase for authentication and real-time data

Solutions Architect, Huawei Feb 2023 – Jan 2024

- Translated business requirements into user stories, API specifications, UML diagrams, and UI prototypes (Figma, Balsamiq) for the Telebirr Super App (30M+ users), acting as primary liaison between business stakeholders, 30+ banking partners, and engineering teams
- Coordinated end-to-end UAT, defect tracking, and release readiness across distributed engineering teams, delivering production releases on schedule

Software Developer, Arifpay Financial Technologies Dec 2021 – Feb 2023

- Rebuilt a socket-based Java data pipeline for banking network payment message exchange (ISO 8583), consolidating 6 bank-specific implementations into a single configurable system supporting multiple banking network standards
- Built a unified abstraction layer over 3-4 POS hardware vendor SDKs, replacing per-vendor codebases with a single API for standardized device communication and eliminating duplicated integration code
- Simplified a multi-threaded Java payment service to a single-threaded model, removing unnecessary thread spawning for a sequential task, cutting code complexity by ~90% and authorization latency from 60s to 2s
- Built Spring Boot REST APIs for a payment gateway across 20+ financial institutions; led 5 engineers in Agile delivery and mentored junior developers

SELECTED PROJECTS

Slack RAG Assistant / Python, AWS Lambda, SQS, DynamoDB, SAM / [GitHub](#)

- Serverless event-driven Slack bot with LLM integration, DynamoDB multi-project routing, and per-channel access control for ASU

PII Filter: LLM Chrome Extension / TypeScript, Chrome Extension APIs / [GitHub](#)

- Enables safe use of public LLMs by automatically intercepting and masking PII in browser prompts before submission, then unmasking sensitive entities in streamed responses, protecting user privacy without disrupting workflow

Privacy-Aware Semantic Retrieval / Python, ChromaDB, Microsoft Presidio, Streamlit / [GitHub](#) | [Live Demo](#)

- Enables private semantic search over personal conversation histories, anonymizes PII at ingestion using Microsoft Presidio, embeds cleaned text into ChromaDB, and surfaces intent-based results

Credit Scoring Predictive Model / Python, PySpark, LightGBM / [GitHub](#)

- Built a full-stack credit scoring platform on a medallion architecture; LightGBM model trained on financial transaction data, served via FastAPI, with a React portal featuring live score updates, explainability, and an OpenAI-powered credit advisor

TECHNICAL SKILLS

AI / LLM: LangChain, CrewAI, MCP, RAG pipelines, ChromaDB, Claude / GPT / Gemini APIs, LightGBM, PySpark ML, Hugging Face

AWS: Lambda, API Gateway, SQS, DynamoDB, S3, ECS/Fargate, Redshift, CDK, SAM, EC2

Languages: Python, Java, SQL, TypeScript / JavaScript

Backend: FastAPI, Spring Boot, REST APIs, Event-Driven Architecture, Docker, Serverless

Data: PostgreSQL, Redshift, DynamoDB, PySpark, SQL (CTEs, window functions, aggregations), Microsoft Presidio